ProveIt research report

Pell growth from divisibility

An affine CRT radix and an explicit 210-variable quartic for Con(ZFC)

Prepared for Vladimir Reshetnikov
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Abstract

We construct a literal quartic polynomial in 210 unrestricted integer variables whose non-solvability is equivalent, provably in ZFC, to the consistency of ZFC, by composing new arithmetic reductions with the repository's identified numerical-calculus calibration. The polynomial is a sum of 150 squares of quadratic polynomials. Its collected expansion has 5,282 nonzero monomials and coefficient height 2^{40}. All coefficients and all 150 residuals are supplied.

The main change replaces an exact exponential-growth request by three quadratic equations forcing the existing growth coordinate to be a sufficiently large Pell ordinate. A normalized Pell permutation modulo powers of two makes this replacement compatible with the odd binomial selector required for CRT. The existing dominant CRT index also provides an affine radix base, removing another coordinate. A delayed sign argument is essential: the new growth ordinate is initially signed. The construction improves the incoming 213-variable quartic in variable count, number of residuals, support and coefficient height. We also provide a 214-variable companion with much smaller coefficients, and an independent 11-positive-variable, degree-174 companion.

These are explicit constructions and ordinary mathematical proofs, with exact finite coefficient and reconstruction checks. No global minimum, historical priority, fresh end-to-end Lean/Rocq build, or proof of consistency is claimed.

Principal equation:

$$P_{210}(x_0,\dots,x_{209}) = \sum_{j=0}^{149} R_j(x_0,\dots,x_{209})^2 = 0, \qquad \boldsymbol{x} \in \mathbb{Z}^{210}.$$

Complete literal residuals: Appendix A.

Executable specification: `equations/principal210/equation.json`.

Offline verification: `python3 -B verify.py`.

Contents

1 The result and its exact scope **1**
1.1 Measured comparisons . . . 1
1.2 What is new and what is inherited . . . 1

2 The inherited arithmetic interface **2**
2.1 The literal local syntax . . . 2

3 Growth from three quadratic equations **3**
3.1 Pell recurrence facts . . . 3

4 The dominant CRT index supplies the radix **4**
4.1 Recovering the signs in the correct order . . . 4
4.2 One binomial bound for the whole front . . . 5

5 A cofinal choice with an odd binomial selector **5**
5.1 The beta moduli need only prime support . . . 6

6 From a zero to a refutation, and back **7**
6.1 Sharp separation of the weighted labels . . . 7
6.2 Soundness of the full arithmetic source . . . 7
6.3 Constructive completeness and the order of choices . . . 7
6.4 The precise logical dependency . . . 8

7 Exact projection, coefficients and reproducibility **8**

8 An independent compact companion **9**
8.1 The complete polynomial . . . 10
8.2 An inverse recurrence with integral coefficients . . . 11
8.3 Soundness in the required order . . . 11
8.4 Positive completeness and exact size . . . 13

9 Further research questions **13**

A The complete literal principal equation **15**

1 The result and its exact scope

The word "simplest" involves several competing measurements. A small number of variables can require a large degree or large fixed coefficients; a quartic with small coefficients can require many variables. This report keeps these measurements separate and makes a concrete improvement in the direct quartic family.

Theorem 1.1 (Explicit principal equation)**.** *Let $R_0, \dots, R_{149} \in \mathbb{Z}[x_0, \dots, x_{209}]$ be the literal quadratic polynomials in Appendix A, and put $P_{210} = \sum_{j=0}^{149} R_j^2$. Then P_{210} has exact degree four, 5,282 collected nonzero monomials and coefficient height $2^{40} = 1,099,511,627,776$. Composing the constructions proved here with the explicitly identified inherited arithmetic and logical calibration gives*

$$\mathsf{ZFC} \vdash \left(\neg\exists \boldsymbol{x} \in \mathbb{Z}^{210}\ P_{210}(\boldsymbol{x}) = 0\right) \iff \mathrm{Con}(\mathsf{ZFC}).$$

All 210 coordinates range independently over the unrestricted integers.

Consistency here means the absence of a finite syntactic refutation in the effective ZFC presentation used by the inherited calibration. The proof is internal to ZFC: proof codes and integer witnesses range over the natural numbers and integers of the theory. It does not introduce a restriction to externally standard codes. We prove the equivalence without assuming either consistency or inconsistency, and we do not prove the polynomial has no zero.

1.1 Measured comparisons

The repository snapshot inspected for this report is `ef0c534c1ae1e312c61a2a9a5a3f18f5516fd42b`. Its incoming archive named `ConZFC_CRT_Index_Anchor_215_2026-10-25.zip` already contains 213- and 214-variable companions. Thus 213, rather than 215, is the relevant existing variable minimum for this comparison. The frozen predecessor is included unchanged under `inherited/anchor/`.

Equation	Variables	Rows	Degree	Monomials	Height
Incoming balanced	215	154	4	5,660	91,679
Incoming companion	214	153	4	5,486	$2^{40} + 76$
Incoming count minimum	213	152	4	5,402	$2.41785\ldots \times 10^{24}$
New balanced companion	214	153	4	5,699	91,679
New principal	**210**	**150**	**4**	**5,282**	$\mathbf{2^{40}}$

The exact height of the incoming 213-variable equation is $2,417,851,639,242,452,488,945,685$. The new principal improves all five reported size quantities except degree, which is unchanged, relative to that equation; it also improves the incoming 214-variable companion in all four non-degree measurements. The balanced 215-variable predecessor still has much smaller coefficients. The new balanced 214-variable companion retains its height while trading 39 additional monomials for one fewer coordinate. It applies only the affine-radix substitution to the original balanced source, retaining that source's original growth certificate. The three-row growth replacement belongs to the principal 210-variable construction.

The maximum coefficient in a principal quadratic residual is 2^{20}. The monomial $x_0^2 x_{209}^2$ has collected coefficient 2^{40}, proving that the reported quartic degree and height are attained. These figures are computed after collecting equal monomials, not from an unexpanded circuit estimate.

1.2 What is new and what is inherited

The use of Pell sequences for Diophantine growth is classical; exponential growth was central to Matiyasevich's work [3]. The contribution here is a compatible replacement inside a specific fully instantiated consistency construction. It consists of a three-row growth certificate, an affine radix

recovered from an existing divisibility, a sign recovery argument, and a cofinal binary-residue construction needed for completeness. The existing common-Hessian and affine projections are reused, with new certificates. Elementary integer coordinate and residual-row operations reduce the final support.

The principal construction works directly with the calibrated numerical calculus $\mathcal{C}_{222}$. It does not simulate K215 and does not instantiate Jones' universal polynomial. The separate compact companion in Section 8 retains its own native/compiler calibration.

2 The inherited arithmetic interface

We describe the semantic coordinates before the exact projections. They are not all separate coordinates after projection; the final equation is defined literally in the appendix.

The positive horizon has the cofinal form

$$t = 16 + 4z^2. \tag{1}$$

The coordinate F provides formula weights $1 + jF$. The quantity $B = \binom{F+t}{t}$ will be recovered from a radix digit, and $N = B^x$ is the generalized CRT modulus. The beta spacing is $y = F(2v + 1)$ and the beta code x satisfies

$$x = F + 1 + (y + 1)r_0, \qquad x = 1 + [1 + (t + 1)y]r_t. \tag{2}$$

Eight independent CRT representatives encode the four local parameters, two pending states, and two beta quotients. The ninth representative C encodes the row index. It is explicitly even and satisfies

$$B\eta = C + F + 1. \tag{3}$$

The shared Pell machinery has its own large center A_* and a square frame containing every actual requested base, exponent and output. Its inherited signed index theorem first authenticates the principal Pell pairs at their nominal, initially signed indices. Its distinguished anchor is the pair for C, and its cap proves

$$C \neq 0, \qquad |k| < |C| \quad \text{for every other retained nominal index } k. \tag{4}$$

After the new growth pair is removed there are 13 retained principal pairs and 15 exact-power requests. These include the requests at base $q + 1$ with indices $F + t$, all eight ordinary representatives, and C; the requests at base q with indices t, $y + 1$, and x; and the two singleton requests with exponent x and bases B and the radix base.

For any integer base b with $|b| \geq 2$, the inherited signed power theorem forces a nonnegative requested exponent k and the exact output b^k. This theorem is used only after the main pairs have been oriented. The frame and anchor theorem does not require the front coordinates to have recovered their arithmetic signs. The new independent Pell witnesses will not be members of this frame; their existing output coordinate F remains a frame atom.

The exact finite changes to this interface are checked by independently reconstructing all 34 retained square-frame atoms, all 13 principal norms and multiplication wires, the whole orientation sum and the anchor cap.

2.1 The literal local syntax

The inherited square-shell constructors are

$$I(a, b) = 2((a + b)^2 + a), \qquad M(a, b) = I(a, b) + 1,$$
$$A(a, b) = 2((a + b)^2 + 3a + 4b + 3), \qquad E(a, b) = A(a, b) + 1.$$

They stand for implication, membership, universal quantification and equality. The numerical truth and falsity codes are 0 and 1. The incoming recursive recoding theorem transports the original numerical calculus to these constructors, including variable labels and the guarded seed schema. This recoding and its finite constructor expressions are retained unchanged.

The source-bound constructor audit gives maximum depth 19 for a local formula label. The nine axiom terms have depths

$$4, 3, 3, 3, 3, 5, 4, 3, 19.$$

Every leaf parameter is strictly below t. The following sharper majorant uses the same expression trees; it does not change the calculus.

Lemma 2.1 (A capacity bound for constructor trees)**.** *Put $d = 2^{19}$ and $L = 8^{d-1}t^d$. Every local formula label is strictly less than L.*

Proof. For $0 \le u, v \le m$, the largest constructor is bounded by $E(m, m)$, and

$$E(m, m) + 1 = 8m^2 + 14m + 8 \le 8(m + 1)^2.$$

A depth-zero leaf has value plus one at most t. The recurrence $H_0 = t$, $H_{r+1} = 8H_r^2$ therefore bounds value plus one at depth r. Its explicit solution is $H_r = 8^{2^r-1}t^{2^r}$. At depth 19 it equals L. Unequal child depths are handled by monotonicity. Thus a label has value at most $L - 1$, as asserted. □

3 Growth from three quadratic equations

The old exact request $F = t^{2^{20}t}$ is stronger than the proof extraction needs. The required interface is a sufficiently large even nonzero F, eventually positive, with a cofinal completeness construction. Introduce three integer coordinates X, Z, u and impose

$$\boxed{Z = tF, \qquad X^2 - 16Z^2 - 8ZF = 1, \qquad F = 2^{19}t(2u + 1).} \tag{5}$$

These are quadratic equations. They replace five coordinates of the old growth request: its principal Pell triple, its modulus and its quotient. The obsolete growth index is removed from the frame. The old principal pair's norm, multiplication wire, orientation term and cap term are removed exactly as specified by the source certificate; F remains a frame atom.

3.1 Pell recurrence facts

For $a \ge 2$ define $X_n(a), Y_n(a)$ by

$$\begin{aligned} X_0 &= 1, & Y_0 &= 0, \\ X_{n+1} &= aX_n + (a^2 - 1)Y_n, & Y_{n+1} &= X_n + aY_n. \end{aligned}$$

The least positive Pell unit at discriminant $a^2 - 1$ is $a + \sqrt{a^2 - 1}$: the ordinate of a nontrivial positive integral solution is at least one, and $(a, 1)$ is a solution. The standard Pell classification therefore gives, for every integer solution with $Y \ne 0$, $|X| = X_n(a)$ and $|Y| = Y_n(a)$ at a unique $n \ge 1$.

We need three elementary consequences:

$$Y_n(a) \equiv n \pmod{a - 1}, \tag{6}$$

$$v_2(Y_n(a)) = v_2(n) \qquad (a \text{ odd}, \ n \ge 1), \tag{7}$$

$$Y_n(a) \ge (2a - 1)^{n-1} \qquad (n \ge 1). \tag{8}$$

The first follows by specializing the integer polynomial recurrence at $a = 1$. For the second, all $X_n(a)$ are odd and $Y_n(a)$ has the parity of n when a is odd; the duplication identity $Y_{2m} = 2X_mY_m$ adds exactly one factor of two at each doubling. For the third, Y_n is increasing and $Y_{n+1} = 2aY_n - Y_{n-1} \ge (2a-1)Y_n$.

Theorem 3.1 (Sign-free growth)**.** *Every integer solution of* (1) *and* (5) *has even nonzero* F *and*

$$|F| \geq L^t, \qquad L = 8^{2^{19}-1}t^{2^{19}}.$$

In particular $|F| \geq t^t > t$.

Proof. The last equation makes F nonzero. Eliminating Z gives

$$X^2 - ((4t+1)^2 - 1)F^2 = 1.$$

Pell classification at the odd center $a = 4t+1$ gives $|F| = Y_n(a)$ for $n \geq 1$. Put $d = 2^{19}$. Since $4t \mid F$, (6) gives $4t \mid n$. Moreover,

$$v_2(n) = v_2(F) = 19 + v_2(t).$$

Consequently the positive integer $n/(4t)$ has exact valuation 17, and $n \geq dt$. Formula (8) gives

$$|F| \geq (8t+1)^{n-1} \geq (8t)^{dt-1} \geq \left(8^{d-1}t^d\right)^t.$$

The last ratio is exactly $8^{t-1}/t \geq 1$. This proves the result with no assumption about the sign of F. □

The flexibility of this certificate is useful. It does not insist that F be one specified power. Section 5 will choose among its Pell ordinates to meet an additional binary-residue condition. Without that choice, merely proving a growth lower bound would not establish completeness.

4 The dominant CRT index supplies the radix

Replace the old base $B[2 + 4(C + \sum A_i)]$ by

$$D = 2(C + F + 1), \qquad q = D^x. \tag{9}$$

The power in this display is still represented by the retained polynomial certificate. It is not an operation in the final polynomial. The definition of D is affine, so its old multiplication coordinate and defining residual can be eliminated without raising degree. Crucially, the actual affine expression is substituted into the main square frame as well.

Equation (3) gives $B \mid D$ directly. Once the signs have been established, the existing anchor supplies $D > 2C$ and bounds every needed binomial upper argument by C. These two roles replace the old explicit factor B and the old sum-majorant.

4.1 Recovering the signs in the correct order

The main shared-Pell frame and C-anchor theorem apply first, while the front coordinates are signed. Theorem 3.1 independently gives even nonzero F with $|F| > t$. Because C is even, $C + F + 1$ is odd and nonzero; thus $|D| \geq 2$ and the request $q = D^x$ forces $x \geq 0$.

Since $y = F(2v+1)$, $|y| \geq |F|$. If $x = 0$, the terminal endpoint in (2) would make $1 + (t+1)y$ divide -1. This is impossible:

$$|1 + (t+1)y| \geq (t+1)|y| - 1 > 1.$$

Hence $x \geq 1$ and $|q| \geq 2$. The retained power at base q and exponent $y+1$ now forces $y + 1 \geq 0$. Negative y would be at most $-|F| < -1$. Therefore $y \geq |F| > 0$.

The value $x = 1$ needs a separate exclusion. The first endpoint would give $-F = (y+1)r_0$, whereas $0 < |F| \leq y < y + 1$. This is impossible. Thus $x \geq 2$, and the terminal endpoint gives

$$x \geq 2 + (t+1)y. \tag{10}$$

It follows that $|q| \geq 2^x \geq 4$ and $|q+1| \geq 3$. The base-$(q+1)$ requests now force all actual CRT representatives and $F+t$ to be nonnegative. If F were negative, then $F+t \leq t-|F| < 0$; hence $F > 0$. The already nonzero C is positive as well. Thus $D > 0$ and $q > 0$.

Finally, if $r_0 \leq -1$, the first endpoint gives $x \leq F - y \leq 0$. Therefore $r_0 \geq 0$ and $x \geq F+1$. We have recovered every required sign without adding a sign witness. In particular, the old assumption of a positive growth output has been replaced by a proved consequence of retained equations.

4.2 One binomial bound for the whole front

By (4), the initial upper argument $F+t$ and all ordinary representatives are below C; the anchor argument itself is C. The lower cutoffs $t, y+1, x$ are at most x. Also

$$D = 2(C+F+1) > 2C, \qquad q = D^x > (2C)^x.$$

For every binomial coefficient in any required low digit, including B,

$$0 \leq \binom{a}{j} \leq C^x < q/2^x \quad (0 \leq a \leq C,\ 0 \leq j \leq k \leq x). \tag{11}$$

Coefficients with $j > a$ are zero; $j = 0$ is included since $C > t \geq 16$. As $x \geq F+1$, this implies $C^x/(q-1) < 1/32$.

For clarity, the retained remainder certificate uses $M = q^x$, denominators q^k, integer ratios $a_k = M/q^k$, and normalized remainders ρ_k. Its exact division equations imply $a_k \mid \rho_k$. Its ternary norm budget

$$M^2 = 1 + 4\sum_k \rho_k^2 + r^2 + s^2 + w(w+1)$$

gives $|\rho_k| < M/2$, hence the actual integer remainder $r_k = \rho_k/a_k$ satisfies $|r_k| < q^k/2$. The genuine binomial low tail is less than $q^k/32$ by (11). Congruence modulo q^k therefore forces the two remainders to agree exactly. This is an equality argument, not an approximation.

The first digit consequently gives $B \equiv \binom{F+t}{t} \pmod q$. Equation (3), with nonzero integer η, also gives

$$|B| \leq C + F + 1 = D/2 < q/2.$$

The true binomial coefficient is positive and below $q/32$, so

$$B = \binom{F+t}{t} > 0. \tag{12}$$

It is at least two. The retained request authenticates $N = B^x$, and

$$q = D^x = (2B\eta)^x = N(2\eta)^x.$$

Thus $N \mid q$, the exact modulus relation required by the generalized range compiler. The same tail argument authenticates all remaining range digits. The inherited local arithmetic bound below 2^x continues to hold: it needs $F \geq t^t$, $t \geq 16$ and (10), all now proved.

5 A cofinal choice with an odd binomial selector

Soundness alone is not enough. The old exact power supplied high powers of two automatically. The new ordinate has a prescribed finite valuation in our completeness construction, so we must choose its other binary digits.

Lemma 5.1 (Pell permutation modulo powers of two)**.** *If $a \equiv 1 \pmod 4$, the map $s \mapsto Y_s(a)$ induces a permutation of $\mathbb{Z}/2^r\mathbb{Z}$ for every $r \geq 1$. Each odd residue has arbitrarily large positive odd indices realizing it.*

Proof. All $X_s(a) \equiv 1 \pmod 4$ and $Y_s(a) \equiv s \pmod 2$. Put $h = 2^r$. Equation (7) gives $v_2(Y_h) = r$. The norm and $8 \mid a^2 - 1$ imply $v_2(X_h^2 - 1) \geq 2r + 3$. Since $X_h + 1$ has valuation one, $X_h \equiv 1 \pmod{2^{2r+2}}$. The addition identity then gives

$$Y_{s+h} - Y_s = (X_h - 1)Y_s + X_s Y_h \equiv 2^r \pmod{2^{r+1}}.$$

The two index lifts s and $s + 2^r$ produce the two distinct output lifts. Induction from modulo two proves bijectivity and periodicity. Positive lifts in the resulting residue class are cofinal and preserve odd parity. □

Theorem 5.2 (Compatible cofinal growth witnesses)**.** *There are cofinally many horizons of the form* (1) *for which* (5) *has arbitrarily large positive solutions F such that $\binom{F+t}{t}$ is odd.*

Proof. Choose the free horizon parameter z divisible by 2^{12}. These horizons remain cofinal and satisfy $t \equiv 16 \pmod{2^{26}}$. Thus $v_2(t) = 4$ and the binary digit of t at position 23 is zero.

Put $d = 2^{19}$, $n_0 = dt$, $y_0 = Y_{n_0}(4t+1)$ and $a_0 = X_{n_0}(4t+1)$. The congruence modulo $4t$ and (7) show that $b_0 = y_0/(dt)$ is an odd integer. Moreover $a_0 \equiv 1 \pmod 4$, since n_0 is even and $X_{2m} = 2X_m^2 - 1$ with X_m odd.

Choose $\ell > 26$ with $2^\ell > t$, and set $r = \ell - 23$. The odd factors $t/16$ and b_0 are units modulo 2^r. By Lemma 5.1, choose a positive odd s, as large as desired, such that

$$(t/16)b_0 Y_s(a_0) \equiv 1 \pmod{2^r}.$$

Set $n = n_0 s$ and

$$F = Y_n(4t+1) = y_0 Y_s(a_0), \quad X = X_n(4t+1), \quad Z = tF, \quad u = \frac{F/(dt) - 1}{2}.$$

Composition of Pell units proves the displayed identity for F, and the odd quotient makes u integral. These values satisfy all three equations and

$$F \equiv 2^{23} \pmod{2^\ell}. \tag{13}$$

Since $t < 2^\ell$ and its bit 23 is zero, the binary supports of F and t are disjoint. There are no carries in their addition. Writing $s_2(m)$ for the binary digit sum, the elementary factorial identity gives

$$v_2\binom{F+t}{t} = s_2(F) + s_2(t) - s_2(F+t) = 0.$$

Thus the binomial is odd. Cofinal positive lifts of s give arbitrarily large F at each chosen horizon. □

The congruence proof is finite arithmetic at an arbitrarily chosen modulus; no analytic inverse theorem or computation of an infinite 2-adic object is needed. The executable modular examples in the archive check this construction at the actual $d = 2^{19}$, but the theorem follows from the recurrence proof for every modulus.

5.1 The beta moduli need only prime support

The new construction has $v_2(F) = 23$ at the selected horizons. We must not reuse the old sufficient condition $t! \mid F(2v+1)$ for arbitrary large t. Fortunately the beta CRT needs only every prime at most t to divide y.

Choose a sufficiently large positive odd w divisible by all odd primes at most t, and put $y = Fw$. The integer y is divisible by every prime at most t. If a prime p divided both $1 + iy$ and $1 + jy$ for $1 \leq i < j \leq t+1$, it would be coprime to y and divide $j - i$. Then $p \leq t$, contradicting $p \mid y$. Thus the beta moduli are pairwise coprime. The odd multiplier can also make y exceed every finite pending state and F, and it has exactly the required form $2v + 1$.

This replacement of an unnecessarily strong factorial divisibility is part of the completeness proof. Keeping the old assertion would leave a real gap.

6 From a zero to a refutation, and back

6.1 Sharp separation of the weighted labels

Lemma 6.1 (Finite-product separation)**.** *Let $L \geq t \geq 1$ and $F \geq L^t$. If two multisets of labels in $[1, L]$ have at most t and $2t+1$ elements respectively and their products of $1 + jF$ are equal, then the multisets are equal.*

Proof. If the factor counts differ, their smaller count r is at most t. The larger product is at least $(F+1)^{r+1}$; the smaller is at most $(1+LF)^r \leq L^r(F+1)^r$. Equality would imply $F+1 \leq L^r \leq F$, a contradiction. The counts therefore agree at some $m \leq t$.

Replace F by an indeterminate T. Each coefficient of either product is a positive integer at most $\binom{m}{k}L^k \leq m^{m-k}L^k \leq L^m \leq F$. For $m = 0$ both products are one. For $m > 0$, corresponding coefficients of the difference have absolute value strictly less than F, including when one of them equals F: the other is positive. A nonzero difference cannot vanish at F, since division by its lowest occurring power of F and reduction modulo F would make a nonzero coefficient of magnitude below F divisible by F. Thus the polynomials agree, and unique factorization in $\mathbb{Q}[T]$ recovers the label multiplicities. □

The decoded local Horn rows have weighted updates

$$S_{i+1}(1 + Fv_i) = S_i(1 + Fp_i)(1 + Fq_i),$$

with missing premises filled by the truth label zero. The endpoint $y = F$ would give an initial beta residue zero and terminal residue one. Positive weights propagate a zero initial state to zero at every row, so this case is impossible. Hence $y > F$, the actual initial state is $F + 1$ and the final state is one. All states are positive. Multiplying the updates and cancelling states gives the equality between the conclusion multiset and the premise multiset plus the target label one. Theorem 3.1 and Lemma 2.1 give precisely the hypotheses of Lemma 6.1.

Cycles in a selected multiset of Horn rules do not cause a gap. Let U be the labels not derivable from the available axioms using those rules. Every selected rule whose conclusion belongs to U has at least one premise in U. If the target one belonged to U, multiplicity balance would say that the number of conclusions in U is one more than the number of premises in U, contradicting the preceding inequality. Thus there is a finite $\mathcal{C}_{222}$ refutation.

6.2 Soundness of the full arithmetic source

From an arbitrary integer zero, first classify the standalone Pell pair to obtain even, nonzero large F. Independently use the retained main frame, anchor and orientation theorem to recover the canonical signed indices. Section 4.1 then proves the front signs. The uniform digit argument recovers B, N, and all range binomials. The inherited generalized binomial-CRT theorem applies because $F \geq L^t \geq t^t$ and hence $F \geq \operatorname{lcm}(1, \ldots, t)^2$. Its two required conclusions are retained: each row has a full prime-power selector exceeding t, and every full prime power of B is assigned to some row. The local arithmetic bound below 2^x converts those congruences into actual local equations.

This decodes the beta states, valid local rules, and their strict parameter ranges. The preceding product and Horn-flow argument yields a finite $\mathcal{C}_{222}$ refutation. All semantic source changes have now been accounted for; the rest of the source decoding is the identified inherited theorem.

6.3 Constructive completeness and the order of choices

Start with a finite $\mathcal{C}_{222}$ refutation. The choices must be made in the following order; this prevents a circular size or congruence argument.

1. Choose a cofinal horizon as in Theorem 5.2, exceeding the refutation length and all finite local parameters; pad by truth steps. Choose positive F and its standalone Pell witnesses with the odd binomial selector property. This fixes $B = \binom{F+t}{t}$.

2. Form the finite weighted pending-state path. Choose the odd multiplier of F so y bounds this path and every prime at most t divides y. Finite CRT chooses an arbitrarily large positive beta code x satisfying the state residues and endpoints. Put $N = B^x$.

3. The generalized binomial CRT supplies the nine representative classes modulo N^2. Choose the eight ordinary representatives positive first. Because N is odd, the ninth class has cofinally many even representatives. Choose C even and larger than every other nominal index and representative. The prescribed index class gives $B \mid C + F + 1$.

4. Set $D = 2(C + F + 1)$ and $q = D^x$ only now. Then $N \mid q$ and $D > 2C$. Evaluate the true requested power outputs, then choose the main square frame and the positive canonical main Pell pairs after all requested atoms have been fixed. The inherited cofinal anchor completion supplies the private comparison witnesses, orientation quotient and cap slack with one fewer principal pair.

5. Take the normalized low remainders. The estimates of Section 4 satisfy the retained common norm budget. Every nonnegative slack has the inherited representation $r^2 + s^2 + w(w + 1)$, supplying the remaining integer witnesses.

This constructs a zero of the changed arithmetic source from every finite refutation. No guessed universal-machine parameter or unknown constant is required.

6.4 The precise logical dependency

The repository's logical endpoints are the named theorems `KEndpoint.c222_iff_zf_refutation` and `ConstructibleChoice.zf_refutation_iff_zfc_refutation`. The first is a theorem about the numerical calculus itself and has no machine argument. The original square-shell recoding transports that calculus without changing finite refutability. The second transfers finite refutations between ZF and ZFC. Combining these with the two arithmetic directions proves Theorem 1.1.

The inherited model argument is internal to the background set theory; it does not replace consistency by the stronger existence of an externally well-founded transitive model. Selected logical files and the exact ordinary proof dependency audit are preserved in the frozen parent. That collection is not a complete compilable Lean dependency closure. This report supplies ordinary proofs formalizable in ZFC and exact finite artifact verification, not a new end-to-end proof-assistant build.

7 Exact projection, coefficients and reproducibility

The unoptimized changed source has 216 coordinates and 156 quadratic rows. Five certified common-Hessian projections and one affine projection give 210 coordinates and 150 rows. Their use here is entirely algebraic.

Lemma 7.1 (Integral common-Hessian projection)**.** *Let Q_i be integer quadratic polynomials and let an integer direction v lie in the kernel of every Hessian. Suppose $v_p = 1$ and some directional linear coefficient ℓ_r is $\epsilon \in \{1, -1\}$. Write $x = y + sv$ with $y_p = 0$. Then $Q_i(x) = Q_i(y) + \ell_i s$, and eliminating $s = -\epsilon Q_r(y)$ gives the quadratic rows $Q_i(y) - \epsilon \ell_i Q_r(y)$ for $i \neq r$. This is a bijection of the integer zero sets, with an explicit integral reconstruction map.*

The proof is the displayed polynomial identity; no rational solvability or divisibility side condition is introduced. The affine projection uses the actual relation

$$4 + 12x_8 + 16x_9 + 4x_{19} - x_{20} + x_{33} = 0$$

at the 211-coordinate stage. It is the difference of two residuals and has unit slope in the eliminated coordinate.

Finally 79 elementary integer operations are applied: coordinate shears or translations with unit diagonal, and additions of integral multiples of one residual to another. Each operation has an integer inverse. Changing the residual basis may change the sum-of-squares polynomial, so polynomial identity with the predecessor is not asserted. Its integer zero set is preserved, which is the required assertion. The independent checker verifies all forward residual identities and their composed inverse.

Stage	Coordinates	Quadratic rows	Collected monomials
Frozen source	219	158	in parent
Standalone Pell growth	217	157	6,943
Affine radix composition	216	156	6,972
Five Hessian projections	211	151	6,926
One affine projection	210	150	6,975
Integral charts and row operations	210	150	5,282

The coordinate and row counts are literal. All final coordinates occur in the collected polynomial. The archived JSON contains each residual as a list of nonzero integer coefficients and sorted coordinate-index multisets. A second JSON contains every collected coefficient of the quartic. The principal equation JSON has SHA-256

```
20a2108353a3756ed7d1ddfdc71695131dcf0b6d94039988d9565ac48e6e1f0a.
```

The supplied verifier reconstructs the source changes independently of the producer, checks the entire main Pell interface, checks the projections, checks all integer operations and the complete expansion, and rejects 22 deliberate corruptions of the direct construction. The printed residual appendix is generated from the frozen final equation and checked back against it. Tests also run with Python's optimization flag, so substantive checks do not depend on `assert` statements.

The commands `python3 -B verify.py` and `python3 -B -O verify.py` use only the Python standard library and need no network. To reconstruct the coefficients in a fresh directory, run

```
python3 -B code/rebuild.py
  -output-dir /tmp/conzfc-210-rebuilt
```

as one command. Finite checks authenticate polynomial coefficients and transformations; the unbounded Pell, CRT and logical arguments are the mathematical proofs above and the explicitly preserved dependencies.

8 An independent compact companion

The direct quartic and the compact construction in this section have different coefficient architectures. Here the coefficients are the complete retained BFS943/radix67 tuple (β, K_0, M_C, M_F), whose largest member has 592884 bits. The new equation has eleven strictly positive integer unknowns, exact degree 174, and 104 live binary arithmetic operations. It retains the native/compiler/refutation calibration. Its full ordinary proof and all coefficient bytes are included in the companion directory. No equality with the direct quartic is asserted.

The inspected source-115 comparator has eleven positive unknowns, degree 211, and 107 operations. Its four coefficient bit lengths agree but its whole-tuple K_0 differs. Thus the comparison concerns domain and size, not an identity between the specialized polynomials. The source-172 equation with ten positive unknowns, degree 214, and 87 operations uses the same complete tuple as the present construction and remains a distinct tradeoff.

8.1 The complete polynomial

All symbols in this section are local. The independent inputs are

$$J, F, \alpha, i, T, w, Y, n, \eta, \zeta, \kappa \in \mathbb{Z}_{>0}.$$

The circuit JSON retains the historical input name `s` for Y; this is a renaming of one input, not an additional variable. Every other symbol below is a polynomial abbreviation. Set $\lambda = 210$ and

$$\begin{aligned}
h &= \beta J, & q &= h + 1,\\
C &= q - F - \alpha, & X &= qw,\\
k &= \eta + \zeta, & c &= kY + \eta,\\
A &= \lambda Y(X+1) + 1, & \Delta &= A^2 - 1,\\
H &= 4A - 5, & v_0 &= XY^2,\\
v &= \lambda^2 v_0, & \ell &= vk,\\
E &= 1 + \ell(\ell + k), & B &= (K_0 + w)C - F.
\end{aligned}$$

$$R = [h(q-F) + \alpha](q^2 - 1) + (M_C + qM_F)J.$$

$$\begin{aligned}
U &= ic^2, & a &= \Delta U^2 E,\\
F_0 &= 1 + a, & G &= EF_0,\\
f &= 1 + 2a, & Q &= 4\Delta a F_0,\\
b &= cT - R, & V &= fb - c,\\
L &= bV, & K &= X(3c - 2X) + 2.
\end{aligned}$$

Define the fixed integer polynomial

$$\begin{aligned}
\mathcal{T}(k, x) = k &- 7350xk(k^2 - 4)\\
&+ 5402250x^2 k(k^2 - 4)(27k^2 + 8)\\
&- 5672362500x^3 k(k^2 - 4)(675k^4 - 324k^2 + 160).
\end{aligned}$$

The final registers are

$$\begin{aligned}
e_* &= \mathcal{T}(k, v_0) - R - 1, & D &= q^7, & M &= hD,\\
Z &= M(n + \kappa - LB) - DB - h(e_* - q^4 Y),\\
P_0 &= Q(Z^2 - M^2 G V^2) + (M^2 G - Z^2), & N &= K - \kappa H.
\end{aligned}$$

The equation is

$$\boxed{P = (P_0 + 1)(3N + 1) - 1 = 0.}$$

There are no extra quantified registers, radicals, variable exponents or division gates. Since P_0, N are integers, $P = 0$ forces both displayed factors to be units. The second factor is 1 modulo 3, and therefore cannot be -1. Consequently

$$P = 0 \quad \Longleftrightarrow \quad P_0 = N = 0.$$

The independently integral projector quotient κ is essential.

8.2 An inverse recurrence with integral coefficients

Let $\psi_b(0) = 0$, $\psi_b(1) = 1$ and $\psi_b(m+2) = 2b\psi_b(m+1) - \psi_b(m)$. Direct recurrence induction gives

$$2\psi_{1+2v}(m) = 2\sum_{j=0}^{m-1} \binom{m+j}{2j+1}(4v)^j.$$

Put

$$\begin{aligned} C_1(k) &= -\frac{k(k^2-4)}{6}, \\ C_2(k) &= \frac{k(k^2-4)(27k^2+8)}{360}, \\ C_3(k) &= -\frac{k(k^2-4)(675k^4-324k^2+160)}{15120}. \end{aligned}$$

These are the first three corrections in the formal inverse $k + C_1(k)v + C_2(k)v^2 + C_3(k)v^3$. With $\lambda = 210$, their denominators clear before any congruence argument:

$$\lambda^2/6 = 7350, \qquad \lambda^4/360 = 5402250, \qquad \lambda^6/15120 = 5672362500.$$

For every integer $m \geq 1$ the following is divisibility in $\mathbb{Z}[x]$:

$$\mathcal{T}(2\psi_{1+2\lambda^2 x}(m), x) - 2m \in x^4\mathbb{Z}[x]. \tag{14}$$

Indeed, the first four recurrence coefficients are

$$2m + 8\lambda^2\binom{m+1}{3}x + 32\lambda^4\binom{m+2}{5}x^2 + 128\lambda^6\binom{m+3}{7}x^3.$$

Substitution makes the coefficients of x, x^2, x^3 vanish as exact polynomials in $\mathbb{Q}[m]$. Both full polynomials have integer coefficients for each integer m, proving (14). The supplied finite certificate checks these polynomial identities; it does not replace the all-index recurrence proof by sampled values.

The transport argument needs a sign bound valid before index recovery. For $k \geq 2$ and $x \geq 1$,

$$\mathcal{T}(k, x) \leq k. \tag{15}$$

To verify it, set $t = k^2 \geq 4$ and $v = \lambda^2 x \geq 1$. The expression subtracted from k is $k(k^2-4)$ times

$$\frac{v}{6} - \frac{v^2(27t+8)}{360} + \frac{v^3(675t^2-324t+160)}{15120}.$$

The last two terms have nonnegative sum because

$$(675t^2 - 324t + 160) - 42(27t+8) = 675(t-4)^2 + 3942(t-4) + 4792 > 0.$$

8.3 Soundness in the required order

The complete proof is retained in `compact11/PROOF_constant_dilation.md`; the key dependencies are given here. At $P_0 = 0$, write $z = Z/M$. Prime valuations in

$$(Q-1)z^2 = G(QV^2 - 1)$$

give positive integral roots $E = \tau^2$, $F_0 = f_0^2$ and $G = g^2$ with $g = \tau f_0$. This uses $Q = 4\Delta^2U^2G$ and $\gcd(E, F_0) = 1$. Adjacent-coefficient Pell descent then gives $y = |z|/g \in \mathbb{Z}_{>0}$, so $z = \pm gy$ is integral. The explicit strong pair has

$$I = 2Ug, \qquad f^2 - \Delta I^2 = 1, \qquad Q = (\Delta I)^2, \qquad c^2 \mid I.$$

The retained signed all-index folding lemma yields $c = \psi_A(p)$ for an odd $p \geq 3$. The first norm yields $k = 2\psi_{1+2v}(m)$. Its index is at least two: $m = 1$ would give $k = 2$ and $c = 2Y + 1$ strictly between the first two positive main Pell ordinates. Since $1 + 2v > A$ and $c > k$, one also has $m < p$.

The free chart no longer has the parent's automatic bound $A > q^2$. A separate argument supplies the needed replacement $c > 2q^4$. If $p \geq 5$, then $c \geq 16A^4 - 12A^2 + 1 > 4A^4 > 4q^4$. If $p = 3$, then $m = 2$ and

$$k = 8\lambda^2 XY^2 + 4, \qquad c = 4A^2 - 1, \qquad \frac{c}{k} > \frac{(X+1)^2}{2X+1} > \frac{X+1}{2}.$$

Together with $c/k < Y + 1$, integrality gives $2Y \geq X$, whence $c > \lambda^2 X^4 - 1 > 2q^4$.

Before the sign of R is known, the uniform auxiliary estimate gives

$$L = bV > 0, \qquad gy < 3L/8, \qquad |b|/q^7 < L/2.$$

For completeness, $b = 0$ would imply $0 < y^2 - c^2 = (c^2 - 1)/(Q - 1) < 1$. The auxiliary Pell index is odd and at least three, so $|V| \geq 4Q - 3$. With $c_3 = 4\Delta + 3 \leq c$ and $f > 2c$,

$$|b| \geq 4\Delta f - \frac{c + c_3}{f} > 4\Delta f - 1 > 4g,$$

while $y < 3|V|/2$. These prove the estimates.

Use the proof-only quantities

$$\sigma = Y/q^3, \quad \rho = e_*/q^7, \quad W = 1 + hL, \quad t = B/h.$$

The carrier identity is

$$z = n + \kappa + \sigma - Wt - \rho.$$

By (15), $e_* \leq k - R - 1 < b$. If $B \leq -1$, the identity gives $n < L/2 - L + |b|/q^7 < 0$. Thus $B \geq 0$ and $C > 0$. The retained mask estimate now gives $3q < R < q^4 - 2$. Hence $c > 2R$, and the normalized strong-rank and signed auxiliary-index lemmas apply to $V \equiv -c \pmod f$ and $V \equiv -R \pmod c$. They give $p = R$.

Integral z implies $q^7 \mid h(e_* - q^4 Y)$. Since $\gcd(q, h) = 1$, $q^4 \mid e_*$. Equation (14), together with $q \mid v_0$, gives $q^4 \mid e_* - (2m - R - 1)$. But $1 \leq m < R < q^4$, and therefore

$$2m = R + 1.$$

No coprimality between q and H has been used.

Write $R = 2r + 1$, so $r \geq 48$. The inherited ratio theorem is uniform for every $\lambda \geq 1$, $Y \geq 1$:

$$\frac{(X+1)^{2r}}{X^r} < \frac{\psi_A(2r+1)}{\psi_{1+2v}(r+1)} < \frac{(X+1)^{2r}}{X^r}\left(1 + \frac{1}{Y(X+1)}\right)^{2r}.$$

It implies $Y > X^r/3$ and $H > X^{r+1}$. The independent equation $N = 0$ gives $H \mid K$; the universal Pell projector then gives

$$H \mid (X - 2^R)(2^R X + 1).$$

The entire product has absolute value below $4 \cdot 4^r X^2 + 8 \cdot 16^r X < X^{r+1} < H$, so it vanishes and $X = 2^R$.

Since $q \mid X$, write $q = 2^a$ with $a \geq 5$. The inequality $R > 3q \geq 7a$ gives $q^7 \mid X$, hence $q^7 \mid v_0$. Equation (14) now gives $q^7 \mid e_*$. Thus ρ is integral, and the carrier gives $Y/q^3 - WB/h \in \mathbb{Z}$. Coprimality and $W \equiv 1 \pmod h$ yield

$$q^3 \mid Y, \qquad h \mid B, \qquad B/h \geq 0.$$

The retained fixed-minus parity, half-binomial rounding and fresh native reconstruction apply. At the entire supplied fixed tuple they produce a finite syntactic ZFC refutation.

8.4 Positive completeness and exact size

Conversely, a refutation supplies the retained canonical outer data, including $X = 2^R$, $R \equiv 3 \pmod 4$, $q^3 \mid Y$ and $B/h \geq 0$. Form the new bases with $\lambda = 210$ and set

$$c = \psi_A(R), \quad k = 2\psi_{1+2v}((R+1)/2), \quad \tau = \chi_{1+2v}((R+1)/2).$$

The uniform ratio theorem gives positive integers $\eta = c - kY$ and $\zeta = k - \eta$. Projection gives positive integral $\kappa = K/H$, while (14) gives integral ρ.

A positive period of the determinant-one Pell matrix modulo $c^2\tau$ provides arbitrarily large pairs (f_0, I_0) with $c^2\tau \mid I_0$. Take $i = I_0/(c^2\tau)$, so $g = \tau f_0$, and choose the period multiple large enough that $g > \kappa + \sigma + |\rho|$. The doubled pair gives $f = 2f_0^2 - 1$, $I = 2f_0 I_0$ and $S = \Delta I$. Put

$$V = \chi_S(R)/S, \qquad y = \psi_S(R), \qquad T = \frac{V + c + Rf}{cf}.$$

The odd quotient identities at $R \equiv 3 \pmod 4$ make T a positive integer and give $V = f(cT - R) - c$. Finally

$$n = gy + (1 + hL)(B/h) + \rho - \kappa - \sigma > 0.$$

These witnesses satisfy both $P_0 = 0$ and $N = 0$. The cofinal choice of the strong period is essential because the new inverse correction may be negative and is larger than the old correction.

The complete highest homogeneous part, with $q_0 = \beta J$ and $k_0 = \eta + \zeta$, is

$$[P]_{174} = -48\lambda^{36} q_0^{47} w^{25} Y^{56} i^{10} k_0^{32} (q_0 - F)^2 (3k_0 Y - 2q_0 w - 4\lambda\kappa Y).$$

In particular,

$$[J^{49} w^{25} Y^{57} i^{10} \eta^{32} \kappa] P = 192\lambda^{37}\beta^{49} > 0.$$

This proves exact degree 174 at the actual fixed coefficients. The independent checker verifies all 104 live gates by exact polynomial cuts, the inverse identity in $\mathbb{Q}[m, x]/(x^4)$, the complete 300-monomial leading form, all four reconstructed coefficient payloads, and rejection of 208 operator mutations. These finite checks support the emitted artifacts; the preceding unbounded argument supplies the zero-existence equivalence.

Deleting κ and restoring K/H inside the shared carrier would suggest a ten-variable degree-176 formula. That shortcut is rejected: fixed λ and free Y do not guarantee $\gcd(q, H) = 1$, so its first-index recovery is not proved.

9 Further research questions

Compress the remaining principal-power interface. Thirteen principal main-Pell pairs remain. The new result shows why a special purpose growth certificate can be cheaper than an exact-power certificate. An analogous specialization of one remaining request is a substantive target. Deleting a pair merely because its nominal index is related to another index would require a proved replacement for all orientation and digit obligations.

Reduce the growth multiplier without increasing degree. The coefficient 2^{19} arises from the depth-19 syntax majorant and a 2-adic index constraint. A shorter equivalent seed term, a tighter source-specific polynomial majorant, or a different congruence forcing a larger index per coefficient could lower height. The current depth argument is sufficient, not a lower bound on the best possible encoding.

Eliminate a coordinate beyond constant Hessian directions. The inherited projections exploit common Hessian kernels and constant residual-row combinations. Nonconstant polynomial combinations, a different Pell chart, or a structural change in the local calculus could expose a new integral projection. Any proposed elimination must retain integer reconstruction and report the resulting support and coefficient growth.

Remove the compact companion's separate projector quotient. The tempting ten-variable shortcut is blocked by a coprimality requirement: the fixed dilation no longer automatically gives $\gcd(q, H) = 1$. A new carrier that proves the necessary denominator separation could improve the compact frontier. Merely substituting a rational quotient into the eleven-variable construction is not justified.

Build a complete formal proof chain. The most valuable verification extension is to formalize the new standalone growth lemma, finite binary-residue lifting, prime-support beta CRT, and changed-source decoding, then compile the full dependency closure through the numerical-calculus/ZF/ZFC endpoints. More bounded examples alone cannot close this formalization boundary.

Specify an optimization criterion. The present 210-variable equation and the 214-variable small-height companion are useful different outcomes. Future searches should state whether the priority is arity at degree four, coefficient height, collected support, binary circuit length, or a weighted combination. No proof here excludes a dramatically smaller equation in any of these senses.

References

[1] V. Reshetnikov, *ProveIt*, inspected snapshot `ef0c534c1ae1e312c61a2a9a5a3f18f5516fd42b`, https://github.com/VladimirReshetnikov/ProveIt. Relevant directories: `Computability/HilbertTenthProblem`, `Computability/DiophantineConsistency/con-zfc-diophantine-forms`, and `docs/incoming`.

[2] *A reused CRT index: an explicit Diophantine consistency equation*, incoming research package, 25 October 2026, `ConZFC_CRT_Index_Anchor_215_2026-10-25.zip`; archive SHA-256 `760428ab91a1f39019ebb452d8b347235b5bb1a0935346e23de78b4542e68677`. Its inherited source snapshot is `953fb4d82d4229f4efc5f9db4cc2bf38ae329bb4`.

[3] Yu. V. Matiyasevich, *Diophantine representation of enumerable predicates*, Izv. Akad. Nauk SSSR Ser. Mat. **35**(1), 3–30 (1971); English translation, Mathematics of the USSR-Izvestiya **5**, 1–28. https://www.mathnet.ru/eng/im1910.

[4] ProveIt entries 115 and 172, compact equation comparison and retained source; exact file paths, byte hashes, coefficient tuple and inherited proofs are included in `compact11/`.

A The complete literal principal equation

Every coordinate $x_0, \dots, x_{209}$ in this appendix is an unrestricted integer. The following 150 displayed residuals define the equation without any omitted coefficient, schematic index, exponentiation request, or external machine parameter:

$$\boxed{P_{210} = R_0^2 + R_1^2 + \cdots + R_{149}^2 = 0.}$$

The sum-of-squares form is itself an explicit polynomial expression. Its 5,282 collected monomials are additionally supplied in `equations/principal210/expanded_polynomial.json`. The final three rows display the new growth block directly: $t = x_0$, $F = x_2$, $X = x_{207}$, $Z = x_{208}$ and $u = x_{209}$.

$$R_0 = x_0 - x_0x_4 - x_4 + x_7$$

$$R_1 = -x_2 - x_4x_5 + x_6x_7$$

$$R_2 = -1 + x_1 - x_6x_7$$

$$R_3 = -2x_2x_{206} - x_3x_{18} + x_4 + 2x_{16}$$

$$\begin{aligned} R_4 = {} & -4 - 3x_8 - x_8^2 - 2x_8x_9 - 4x_9 - x_9^2 \\ & + x_{19} \end{aligned}$$

$$\begin{aligned} R_5 = {} & x_8 - x_8^2 - 8x_8x_{19} - 4x_8x_{20} - 16x_{19}^2 - 16x_{19}x_{20} \\ & - 4x_{20}^2 + x_{21} \end{aligned}$$

$$R_6 = 1 + 2x_{21} - 4x_{21}^2 + x_{22}$$

$$R_7 = -2 - x_9^2 - 2x_9x_{10} - x_{10}^2 + x_{23}$$

$$R_8 = 1 - x_8^2 - 2x_8x_{10} - x_{10}^2 + x_{24}$$

$$\begin{aligned} R_9 = {} & -29 - 61x_8 - 35x_8^2 - 96x_8x_{10} - 24x_8x_{19} - 4x_8x_{23} \\ & - 24x_8x_{24} + 16x_9 - 2x_9^2 - 8x_9x_{23} - 96x_{10} - 64x_{10}^2 \\ & - 32x_{10}x_{19} - 32x_{10}x_{24} - 28x_{19} - 4x_{19}^2 - 8x_{19}x_{24} + 8x_{23} \\ & - 4x_{23}^2 - 24x_{24} - 4x_{24}^2 + x_{25} \end{aligned}$$

$$\begin{aligned} R_{10} = {} & -4 + 10x_8 - 2x_8^2 - 8x_8x_9 - 8x_8x_{23} - 4x_{25}^2 \\ & + x_{26} - 2x_{43} \end{aligned}$$

$$\begin{aligned} R_{11} = {} & -4 - 16x_{10}^2 - 16x_{10}x_{23} - 16x_{10}x_{24} - 4x_{23}^2 - 8x_{23}x_{24} \\ & - 4x_{24}^2 + x_{27} \end{aligned}$$

$$\begin{aligned} R_{12} = {} & -7 + 24x_{10} - 16x_{10}^2 - 16x_{10}x_{19} - 32x_{10}x_{24} - 16x_{10}x_{27} \\ & + 10x_{19} - 4x_{19}^2 - 16x_{19}x_{24} - 8x_{19}x_{27} + 24x_{24} - 16x_{24}^2 \\ & - 16x_{24}x_{27} + 12x_{27} - 4x_{27}^2 + x_{28} \end{aligned}$$

$$\begin{aligned}R_{13} = &-114 - 127x_8 - 37x_8^2 - 24x_8x_9 - 112x_8x_{10} - 4x_8x_{23}\\ &- 24x_8x_{24} - 67x_9 - 37x_9^2 - 112x_9x_{10} - 24x_9x_{23} - 4x_9x_{24}\\ &- 224x_{10} - 128x_{10}^2 - 32x_{10}x_{23} - 32x_{10}x_{24} - 16x_{23} - 4x_{23}^2\\ &- 40x_{24} - 4x_{24}^2 + x_{29}\end{aligned}$$

$$\begin{aligned}R_{14} = &-30 - 14x_8 - 2x_8^2 - 24x_8x_9 - 32x_8x_{10} - 8x_8x_{23}\\ &- 96x_9 - 72x_9^2 - 192x_9x_{10} - 48x_9x_{23} - 128x_{10} - 128x_{10}^2\\ &- 64x_{10}x_{23} - 32x_{23} - 8x_{23}^2 - 4x_{29}^2 + x_{30}\end{aligned}$$

$$\begin{aligned}R_{15} = &-74 - 42x_8 + 8x_8x_{10} + 16x_8x_{23} - 84x_9 + 16x_9x_{10}\\ &+ 32x_9x_{23} + 29x_{10} - 3x_{10}^2 - 4x_{10}x_{19} - 8x_{10}x_{23} + 21x_{19}\\ &- 8x_{19}x_{23} + 40x_{23} - 4x_{23}^2 + x_{31} - x_{59}\end{aligned}$$

$$\begin{aligned}R_{16} = &-131 - 184x_8 - 64x_8^2 - 160x_8x_{10} - 64x_8x_{23} - 32x_8x_{24}\\ &- 32x_8x_{31} - 226x_{10} - 100x_{10}^2 - 80x_{10}x_{23} - 40x_{10}x_{24} - 40x_{10}x_{31}\\ &- 88x_{23} - 16x_{23}^2 - 16x_{23}x_{24} - 16x_{23}x_{31} - 46x_{24} - 4x_{24}^2\\ &- 8x_{24}x_{31} - 44x_{31} - 4x_{31}^2 + x_{32}\end{aligned}$$

$$R_{17} = -4x_{19}^2 + x_{20}$$

$$\begin{aligned}R_{18} = &\,x_8 - x_8^2 + 8x_8x_{19} - 4x_8x_{20} - 16x_{19}^2 + 16x_{19}x_{20}\\ &- 4x_{20}^2 + x_{33}\end{aligned}$$

$$R_{19} = 1 + 2x_{33} - 4x_{33}^2 + x_{34}$$

$$R_{20} = -x_9^2 - 4x_9x_{34} - 4x_{34}^2 + x_{35}$$

$$\begin{aligned}R_{21} = &-44 - 52x_8 - 16x_8^2 - 64x_8x_9 + 32x_8x_{19} - 104x_9\\ &- 64x_9^2 + 64x_9x_{19} + 50x_{19} - 16x_{19}^2 + x_{36}\end{aligned}$$

$$\begin{aligned}R_{22} = &\,8x_9 - 62x_9^2 - 92x_9x_{10} - 32x_9x_{23} - 32x_9x_{36} + 6x_{10}\\ &- 34x_{10}^2 - 24x_{10}x_{23} - 24x_{10}x_{36} - 4x_{23}^2 - 8x_{23}x_{36} + 4x_{36}\\ &- 4x_{36}^2 + x_{37}\end{aligned}$$

$$\begin{aligned}R_{23} = &-90 - 152x_8 - 64x_8^2 - 96x_8x_{10} - 32x_8x_{24} - 32x_8x_{37}\\ &- 114x_{10} - 36x_{10}^2 - 24x_{10}x_{24} - 24x_{10}x_{37} - 38x_{24} - 4x_{24}^2\\ &- 8x_{24}x_{37} - 36x_{37} - 4x_{37}^2 + x_{38}\end{aligned}$$

$$R_{24} = -x_{10}^2 - 2x_{10}x_{11} - x_{11}^2 + x_{39}$$

$$\begin{aligned}R_{25} = &-7 + 8x_8 - 3x_8^2 - 8x_8x_9 - 8x_8x_{23} - 8x_8x_{24}\\ &- 4x_8x_{39} - 8x_9x_{24} - 2x_{10}^2 - 8x_{10}x_{39} - 8x_{23}x_{24} + 6x_{24}\\ &- 4x_{24}^2 - 4x_{39}^2 + x_{40} - x_{43}\end{aligned}$$

$$\begin{aligned}R_{26} = &-16x_8^2 - 36x_8x_{10} - 32x_8x_{24} - 40x_8x_{39} - 16x_8x_{40} - 4x_{10}\\ &- 22x_{10}^2 - 32x_{10}x_{24} - 48x_{10}x_{39} - 16x_{10}x_{40} - 2x_{24} - 16x_{24}^2\\ &- 32x_{24}x_{39} - 16x_{24}x_{40} - 4x_{39} - 24x_{39}^2 - 16x_{39}x_{40} - 4x_{40}^2\\ &+ x_{41}\end{aligned}$$

$$R_{27} = x_{10} - 25x_{10}^2 - 40x_{10}x_{24} - 20x_{10}x_{27} - 16x_{24}^2 - 16x_{24}x_{27} - 4x_{27}^2 + x_{42}$$

$$R_{28} = -1 + 8x_9 - 4x_9^2 - 8x_9x_{23} + 8x_{23} - 4x_{23}^2 + x_{43}$$

$$R_{29} = -1 - 16x_9^2 - 8x_9x_{10} - 32x_9x_{23} - 16x_9x_{43} - x_{10}^2 - 8x_{10}x_{23} - 4x_{10}x_{43} - 16x_{23}^2 - 16x_{23}x_{43} - 4x_{43}^2 + x_{44}$$

$$R_{30} = 3 + x_9 - 2x_9x_{10} - 4x_9x_{44} - 3x_{10}^2 - 8x_{10}x_{44} - x_{23} - 4x_{44}^2 + x_{45}$$

$$R_{31} = -1 - 4x_{45} - 4x_{45}^2 + x_{46}$$

$$R_{32} = -1 - x_{10}^2 - 8x_{10}x_{45} - 4x_{10}x_{46} - 16x_{45}^2 - 16x_{45}x_{46} - 4x_{46}^2 + x_{47}$$

$$R_{33} = 2 + x_9 - x_9^2 - 4x_9x_{10} - 4x_9x_{47} - 4x_{10}^2 - 8x_{10}x_{47} - 4x_{47}^2 + x_{48}$$

$$R_{34} = 1 - 4x_{42}^2 - 8x_{42}x_{48} - 4x_{48}^2 + x_{49}$$

$$R_{35} = 1 + x_8 - x_8^2 - 8x_8x_{42} - 4x_8x_{49} - 16x_{42}^2 - 16x_{42}x_{49} - 4x_{49}^2 + x_{50}$$

$$R_{36} = -1 - 4x_{50} - 4x_{50}^2 + x_{51}$$

$$R_{37} = -1 - x_{10}^2 - 8x_{10}x_{50} - 4x_{10}x_{51} - 16x_{50}^2 - 16x_{50}x_{51} - 4x_{51}^2 + x_{52}$$

$$R_{38} = 2 + x_8 - x_8^2 - 4x_8x_{10} - 4x_8x_{52} - 4x_{10}^2 - 8x_{10}x_{52} - 4x_{52}^2 + x_{53}$$

$$R_{39} = -10 - 12x_{53} - 4x_{53}^2 + x_{54}$$

$$R_{40} = -19 + 4x_9 - 4x_9^2 - 8x_9x_{23} - 16x_9x_{53} - 8x_9x_{54} + 4x_{23} - 4x_{23}^2 - 16x_{23}x_{53} - 8x_{23}x_{54} - 16x_{53} - 24x_{53}^2 - 16x_{53}x_{54} + 6x_{54} - 4x_{54}^2 + x_{55}$$

$$R_{41} = -10 + x_9 - 25x_9^2 - 40x_9x_{23} - 20x_9x_{55} - 16x_{23}^2 - 16x_{23}x_{55} - 12x_{48} - 4x_{48}^2 - 4x_{55}^2 + x_{56}$$

$$R_{42} = 21 + 24x_{48} - 8x_{48}^2 - 16x_{48}x_{56} - 2x_{56} - 4x_{56}^2 + x_{57}$$

$$R_{43} = -1 + 4x_8 + 8x_9 + x_{10} - x_{10}^2 - 4x_{10}x_{57} - 2x_{19} - 4x_{57}^2 + x_{58} + 12x_{59} - 4x_{59}^2$$

$$\begin{aligned} R_{44} = & -22 - 38x_8 - 16x_8^2 - 64x_8x_9 + 8x_8x_{10} + 16x_8x_{19} \\ & - 76x_9 - 64x_9^2 + 16x_9x_{10} + 32x_9x_{19} + 11x_{10} - x_{10}^2 \\ & - 4x_{10}x_{19} + 19x_{19} - 4x_{19}^2 + x_{59} \end{aligned}$$

$$\begin{aligned} R_{45} = & -4 + 8x_8 + 16x_9 - 4x_{19} - 4x_{58}^2 + 24x_{59} \\ & - 8x_{59}^2 + x_{60} \end{aligned}$$

$$R_{46} = -1 - 4x_{60} - 4x_{60}^2 + x_{61}$$

$$R_{47} = -x_2x_{13} + x_{62}$$

$$R_{48} = x_{12} - x_{13} - 2x_{22}x_{62} - x_{63}$$

$$R_{49} = x_{12} - x_{13} - 2x_{26}x_{62} - x_{64}$$

$$R_{50} = x_{12} - x_{13} - 2x_{28}x_{62} - 2x_{62} - x_{65}$$

$$R_{51} = x_{12} - x_{13} - 2x_{30}x_{62} - x_{66}$$

$$R_{52} = x_{12} - x_{13} - 2x_{32}x_{62} + 2x_{62} - x_{67}$$

$$R_{53} = x_{12} - x_{13} - 4x_{34}x_{62} - 2x_{35}x_{62} - x_{68}$$

$$R_{54} = x_{12} - x_{13} - 2x_{38}x_{62} - x_{69}$$

$$R_{55} = 2x_{28}x_{62} - 2x_{41}x_{62} - 2x_{62} + x_{65} - x_{70}$$

$$R_{56} = -x_8x_9 + x_8x_{10} + x_9x_{10} - x_{10}^2 + x_{71}$$

$$R_{57} = 1 + 2x_{11} + x_{11}^2 - x_{71}^2 + x_{72}$$

$$\begin{aligned} R_{58} = & -x_{12}^2 + 2x_{12}x_{13} + 2x_{12}x_{74} - x_{13}^2 - 2x_{13}x_{74} - x_{72}^2 \\ & + x_{73} - x_{74}^2 \end{aligned}$$

$$R_{59} = -4x_{60}x_{62} - 2x_{61}x_{62} + x_{74}$$

$$R_{60} = -x_9x_{62} - x_{13} + x_{75}x_{76} + x_{77}$$

$$R_{61} = -1 + 8x_2 + 4x_2x_8 + 8x_2x_9 - 2x_2x_{19} + x_{75}$$

$$R_{62} = -x_8x_{201} - x_{12} + x_{76}$$

$$R_{63} = x_9x_{201} - 2x_{19}x_{62} + 2x_{32}x_{62} + x_{67} + x_{78}$$

$$R_{64} = -x_{12}x_{63} + x_{13}x_{63} + x_{79}$$

$$R_{65} = -x_{64}x_{79} + x_{80}$$

$$R_{66} = -x_{65}x_{80} + x_{81}$$

$$R_{67} = -x_{66}x_{81} + x_{82}$$

$$R_{68} = -x_{67}x_{82} + x_{83}$$

$$R_{69} = -x_{68}x_{83} + x_{84}$$

$$R_{70} = -x_{69}x_{84} + x_{85}$$

$$R_{71} = -x_{70}x_{85} + x_{86}$$

$$R_{72} = -x_{77}x_{86} + x_{87}$$

$$R_{73} = -x_{78}x_{87} + x_{88}$$

$$R_{74} = -x_{88}^2 + x_{89}$$

$$R_{75} = -x_4 - 2x_4x_{16} + 2x_{16} + x_{90}$$

$$R_{76} = -x_{14}x_{90} + x_{91}$$

$$\begin{aligned} R_{77} = & -2x_1^2 + 2x_1x_{12} + 2x_1x_{13} + 2x_1x_{91} + 2x_1x_{92} - x_{12}^2 \\ & - 2x_{12}x_{91} - x_{13}^2 - 2x_{13}x_{92} + x_{17}x_{93} - x_{73}x_{89} - x_{91}^2 \\ & - x_{92}^2 \end{aligned}$$

$$R_{78} = -x_4x_{15} + x_{15} - x_{15}x_{90} + x_{92}$$

$$R_{79} = -x_{94}x_{97} + x_{97} + x_{99}$$

$$R_{80} = -x_3x_{120} + x_{95}x_{193} - x_{98} - x_{99}x_{120}$$

$$R_{81} = x_{100}x_{193} - x_{101}x_{123} - x_{102}$$

$$R_{82} = x_{103}x_{193} - x_{104}x_{123} - x_{105}$$

$$R_{83} = x_{106}x_{193} - x_{107}x_{123} - x_{108}$$

$$R_{84} = x_{109}x_{193} - x_{110}x_{123} - x_{111}$$

$$R_{85} = x_{112}x_{194} - x_{114}x_{123} - x_{115}$$

$$R_{86} = x_{116}x_{194} - x_{117}x_{123} - x_{118}$$

$$R_{87} = -x_{17}x_{120} + x_{123}$$

$$R_{88} = x_{119} - x_{121}x_{123} - x_{122}$$

$$R_{89} = -x_{123}x_{125} + x_{124} - x_{126}$$

$$R_{90} = -x_{123}x_{128} + x_{127}x_{193} - x_{129}$$

$$R_{91} = x_{130} - 2x_{130}x_{132} + x_{133}$$

$$R_{92} = -x_{130}x_{133} - x_{133} + x_{134}$$

$$R_{93} = -1 + x_{131}^2 + x_{133}^2 - x_{134}^2$$

$$R_{94} = 1 - x_{131}x_{136} - x_{135} + x_{137}$$

$$R_{95} = -1 + 2x_{135} - x_{135}^2 + 2x_{135}x_{137} + x_{136}^2$$

$$R_{96} = 1 + x_3^2 - 2x_3x_{131} + x_{138}$$

$$R_{97} = -x_3x_{170} + x_{17} + x_{138}x_{139} - x_{169}$$

$$\begin{aligned} R_{98} = &-5 - 8x_2 - 4x_2^2 - 16x_2x_{16} + 4x_2x_{131} - 16x_{16} \\ &- 16x_{16}^2 + 8x_{16}x_{131} + 4x_{131} - x_{140} \end{aligned}$$

$$\begin{aligned} R_{99} = &-1 - 2x_2x_{170} - 4x_{16}x_{170} + x_{94} + x_{140}x_{141} - x_{169} \\ &- 2x_{170} \end{aligned}$$

$$R_{100} = -x_{131}x_{143} - x_{142} + x_{144}$$

$$R_{101} = -1 - x_{142}^2 + 2x_{142}x_{144} + x_{143}^2$$

$$R_{102} = -x_{131}x_{146} - x_{145} + x_{147}$$

$$R_{103} = -1 - x_{145}^2 + 2x_{145}x_{147} + x_{146}^2$$

$$R_{104} = -x_{131}x_{149} - x_{148} + x_{150}$$

$$R_{105} = -1 - x_{148}^2 + 2x_{148}x_{150} + x_{149}^2$$

$$R_{106} = -x_{131}x_{152} - x_{151} + x_{153}$$

$$R_{107} = -1 - x_{151}^2 + 2x_{151}x_{153} + x_{152}^2$$

$$R_{108} = -x_{131}x_{155} - x_{154} + x_{156}$$

$$R_{109} = -1 - x_{154}^2 + 2x_{154}x_{156} + x_{155}^2$$

$$R_{110} = -x_{131}x_{158} - x_{157} + x_{159}$$

$$R_{111} = -1 - x_{157}^2 + 2x_{157}x_{159} + x_{158}^2$$

$$R_{112} = 1 - x_{131}x_{161} - x_{160} + x_{162}$$

$$R_{113} = -1 + 2x_{160} - x_{160}^2 + 2x_{160}x_{162} + x_{161}^2$$

$$R_{114} = -x_{131}x_{164} - x_{163} + x_{165}$$

$$R_{115} = -1 - x_{163}^2 + 2x_{163}x_{165} + x_{164}^2$$

$$R_{116} = -x_{131}x_{167} - x_{166} + x_{168}$$

$$R_{117} = -1 - x_{166}^2 + 2x_{166}x_{168} + x_{167}^2$$

$$R_{118} = 1 - x_{131}x_{170} - x_{169} + x_{171}$$

$$R_{119} = -1 + 2x_{169} - x_{169}^2 + 2x_{169}x_{171} + x_{170}^2$$

$$R_{120} = -x_{131}x_{173} - x_{172} + x_{174}$$

$$R_{121} = -1 - x_{172}^2 + 2x_{172}x_{174} + x_{173}^2$$

$$R_{122} = -x_{131}x_{176} - x_{175} + x_{177}$$

$$R_{123} = -1 - x_{175}^2 + 2x_{175}x_{177} + x_{176}^2$$

$$\begin{aligned} R_{124} = & -8 - 2x_0 - 4x_0^2 - 2x_0x_2 - x_1^2 - 8x_2 \\ & - 6x_2^2 - 16x_2x_{16} - x_3^2 - x_4^2 - x_8^2 - x_9^2 \\ & - x_{10}^2 - x_{11}^2 - x_{12}^2 - x_{13}^2 - x_{14}^2 - x_{15}^2 \\ & - 16x_{16} - 20x_{16}^2 - x_{17}^2 + 2x_{94} - 2x_{94}^2 - x_{95}^2 \\ & - x_{96}^2 - x_{100}^2 - x_{103}^2 - x_{106}^2 - x_{109}^2 - x_{112}^2 \\ & - x_{113}^2 - x_{116}^2 - x_{119}^2 - x_{120}^2 - x_{124}^2 - x_{127}^2 \\ & + x_{130} \end{aligned}$$

$$R_{125} = -16 + x_0 - 4x_{178}^2$$

$$R_{126} = -x_{131}x_{180} + x_{189}$$

$$R_{127} = -1 + x_{179}^2 + x_{180}^2 - x_{189}^2$$

$$R_{128} = -x_{182}x_{183} + x_{190}$$

$$R_{129} = -1 + x_{181}^2 + x_{182}^2 - x_{190}^2$$

$$R_{130} = -x_{176}^2 + x_{191}$$

$$R_{131} = x_{180} - 2x_{185}x_{191} - x_{191}$$

$$R_{132} = -1 - 4x_{180} - 8x_{180}x_{184} + x_{183}$$

$$R_{133} = -x_{131} - x_{179}x_{186} + x_{183}$$

$$R_{134} = -x_{177} - x_{179}x_{187} + x_{181}$$

$$R_{135} = -2x_{16} - 4x_{176}x_{188} + x_{182}$$

$$\begin{aligned} R_{136} = 9 &+ 2x_0^2 + 2x_0x_2 - 2x_0x_{136} - 2x_0x_{143} + x_1^2 \\ &- 2x_1x_{170} + x_2^2 - 2x_2x_{143} + x_4^2 - 2x_4x_{161} + x_8^2 \\ &- 2x_8x_{146} + x_9^2 - 2x_9x_{149} + x_{10}^2 - 2x_{10}x_{152} + x_{11}^2 \\ &- 2x_{11}x_{155} + x_{12}^2 - 2x_{12}x_{158} + x_{13}^2 - 2x_{13}x_{164} + x_{14}^2 \\ &- 2x_{14}x_{167} + x_{15}^2 - 2x_{15}x_{173} - 4x_{16}x_{176} - x_{131}x_{192} + x_{137}^2 \\ &- 2x_{144} + x_{144}^2 - 2x_{147} + x_{147}^2 - 2x_{150} + x_{150}^2 \\ &- 2x_{153} + x_{153}^2 - 2x_{156} + x_{156}^2 - 2x_{159} + x_{159}^2 \\ &+ x_{162}^2 - 2x_{165} + x_{165}^2 - 2x_{168} + x_{168}^2 + x_{171}^2 \\ &- 2x_{174} + x_{174}^2 + 2x_{176}^2 - 2x_{177} + x_{177}^2 + x_{192} \\ &- x_{195}^2 - x_{196}^2 - x_{197} - x_{197}^2 \end{aligned}$$

$$R_{137} = -x_{96}x_{193} + x_{120}$$

$$R_{138} = -x_{113}x_{194} + x_{120}$$

$$\begin{aligned} R_{139} = &-1 - 4x_{16}^2 - x_{136}^2 - x_{143}^2 - x_{146}^2 - x_{149}^2 \\ &- x_{152}^2 - x_{155}^2 - x_{158}^2 - x_{161}^2 - x_{164}^2 - x_{167}^2 \\ &- x_{170}^2 - x_{173}^2 + x_{176}^2 - x_{195}^2 - x_{196}^2 - x_{197} \\ &- x_{197}^2 \end{aligned}$$

$$\begin{aligned} R_{140} = &-3 - 4x_{98}^2 - 4x_{102}^2 - 4x_{105}^2 - 4x_{108}^2 - 4x_{111}^2 \\ &- 4x_{115}^2 - 4x_{118}^2 + x_{120}^2 - 4x_{122}^2 - 4x_{126}^2 - 4x_{129}^2 \\ &- x_{198}^2 - x_{199}^2 + 3x_{200} - x_{200}^2 \end{aligned}$$

$$R_{141} = -x_2x_{12} + x_{201}$$

$$\begin{aligned} R_{142} = x_{94}x_{203} &- 2x_{95}x_{143} - 2x_{100}x_{146} - 2x_{103}x_{149} - 2x_{106}x_{152} - 2x_{109}x_{155} \\ &- 2x_{112}x_{158} - 2x_{116}x_{164} - 2x_{119}x_{167} - 2x_{124}x_{173} - 2x_{127}x_{176} - 2x_{131}x_{203} \\ &+ 2x_{143}x_{144} + 2x_{146}x_{147} + 2x_{149}x_{150} + 2x_{152}x_{153} + 2x_{155}x_{156} + 2x_{158}x_{159} \\ &+ 2x_{164}x_{165} + 2x_{167}x_{168} + 2x_{173}x_{174} + 2x_{176}x_{177} + x_{202} \end{aligned}$$

$$\begin{aligned} R_{143} = &-10 - x_{94}x_{202} + x_{95}^2 - 2x_{95}x_{142} + x_{100}^2 - 2x_{100}x_{145} \\ &+ x_{103}^2 - 2x_{103}x_{148} + x_{106}^2 - 2x_{106}x_{151} + x_{109}^2 - 2x_{109}x_{154} \\ &+ x_{112}^2 - 2x_{112}x_{157} + x_{116}^2 - 2x_{116}x_{163} + x_{119}^2 - 2x_{119}x_{166} \\ &+ x_{124}^2 - 2x_{124}x_{172} + x_{127}^2 - 2x_{127}x_{175} + 2x_{142}x_{144} + 2x_{145}x_{147} \\ &+ 2x_{148}x_{150} + 2x_{151}x_{153} + 2x_{154}x_{156} + 2x_{157}x_{159} + 2x_{163}x_{165} + 2x_{166}x_{168} \\ &+ 2x_{172}x_{174} + 2x_{175}x_{177} + x_{203} \end{aligned}$$

$$\begin{aligned} R_{144} = x_{94}x_{205} &- 2x_{96}x_{136} - 2x_{113}x_{161} - 2x_{120}x_{170} - 2x_{131}x_{205} + 2x_{136} \\ &+ 2x_{136}x_{137} + 2x_{161} + 2x_{161}x_{162} + 2x_{170} + 2x_{170}x_{171} + x_{204} \\ &- x_{205} \end{aligned}$$

$$\begin{aligned} R_{145} = & -2 - x_{94}x_{204} + x_{96}^2 - 2x_{96}x_{135} + x_{113}^2 - 2x_{113}x_{160} \\ & + x_{120}^2 - 2x_{120}x_{169} + 2x_{135} + 2x_{135}x_{137} + 2x_{160} + 2x_{160}x_{162} \\ & + x_{169}^2 - x_{170}^2 + x_{204} + x_{205} \end{aligned}$$

$$R_{146} = -1 - x_2 - 2x_2x_{206} + x_4$$

$$R_{147} = -x_0x_2 + x_{208}$$

$$R_{148} = -1 - 8x_2x_{208} + x_{207}^2 - 16x_{208}^2$$

$$R_{149} = -524288x_0 - 1048576x_0x_{209} + x_2$$